Surface Form or Content? Decomposing Separability in AI-Generated Text Detection Benchmarks

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***Abstract*—A detector that scores 99% on a machine-generated-text benchmark may have modelled how machine language differs from human language, or how one corpus happened to be punctuated and encoded, and the headline number does not distinguish the two. We propose a decomposition that measures the difference and apply it to two widely used corpora. A surface-only model reads 47 orthographic features and never sees a word, a content-only model reads text with punctuation, casing and non-ASCII characters stripped away, and a full model is a fine-tuned transformer on raw text. Across DAIGT V2 and HC3, five model families and eight configurations per corpus, the two benchmarks separate. On HC3 the surface and content arms are indistinguishable under a paired test, 3.20% against 3.26% error, so orthography alone matches an unfiltered bag-of-words over the whole vocabulary, and the null holds on all five of five independent partitions with the sign changing between them. On DAIGT V2 content is stronger by a factor of 7.9 in error. Closing the document-length channel both arms share widens the DAIGT V2 gap to 10.6 times and turns HC3's parity into a 2.91-point advantage for orthography. A SHAP attribution locates the asymmetry, since one feature, the rate of spaces preceding punctuation, dominates HC3 at 2.3 times the next while no orthographic feature dominates DAIGT V2. Three controls change what the remaining results mean. BERT emits identical token identifiers for strings differing only by that whitespace cue, so it cannot represent the cue and still reaches 0.9916 weighted F1, making the cue sufficient in isolation but unnecessary in practice. A pipeline control confirms this null is not a cleaning step that failed to execute. A label-free control shows these models assign the human label to uniform random character strings in 100% of cases at 0.98 mean confidence, so an adversarial collapse to 0.3644 weighted F1 cannot be read as a targeted vulnerability. Surface form is informative on both corpora, and we claim only that it reaches parity with content on one.**

***Index Terms*—AI-generated text detection, benchmark evaluation, surface features, shortcut learning, tokenisation, SHAP, paired significance testing**

I. Introduction

Detectors of machine-generated text report accuracy at or above 97% on the benchmarks the field uses most, and frequently above 99% [1], [2], [3], [4]. Read directly, those numbers say the task is close to solved on the domains these corpora represent. Read carefully, they say something weaker, because a benchmark score measures the separability of a particular corpus and not the capability a reader infers from it.

The difference matters because separability has more than one source. A detector can reach a high score by modelling how machine-generated language differs from human language, or by modelling how one corpus was assembled, punctuated and encoded. Both produce the same headline figure. Individual instances of the second have been documented, including a whitespace convention in HC3 [5], a length confound in the same corpus [6], punctuation inconsistencies requiring standardisation before comparison [7], and prompt-specific collection shortcuts [8]. Each of those is a report of one artefact. What is missing is a way to ask how much, in total, a given benchmark's separability owes to surface form rather than content, and to place two benchmarks on the same axis so they can be compared.

This paper proposes that measurement and applies it. We compare three arms on identical splits. A *surface-only* model reads 47 orthographic features and never sees a word. A *content-only* model reads text after punctuation, casing and non-ASCII characters have been stripped away. A *full* model is a fine-tuned transformer on raw text. The gap between the first two bounds how much of a benchmark a detector could pass without reading language at all. Because the first two arms share a classifier family, they are directly comparable to each other, and the third is reported as a reference point rather than as a matched arm.

Applied to DAIGT V2 and HC3 across five model families, the answer differs sharply between the two corpora. On HC3 the surface-only and content-only arms are indistinguishable, 3.20% against 3.26% test error, and a paired test over 10,732 documents cannot separate them. On DAIGT V2 content is stronger by a factor of 7.9 in error. The same contrast appears in the model comparison, where the best classical configuration falls 0.0007 weighted F1 short of the best transformer on DAIGT V2 and short by an error-rate factor of 16 on HC3. That tie is a statement about text budget rather than architecture, and we measure it rather than assume it, since refitting the classical grid on the exact span each transformer's tokeniser kept removes it and the transformer then leads on both corpora.

We also report two negative results, because each corrects a claim that an uncontrolled version of this analysis had reached and that the surrounding literature invites. The whitespace cue that dominates discussion of HC3 turns out to be sufficient in isolation yet unnecessary in practice, since BERT's tokeniser provably cannot represent it and BERT still reaches 0.9916 weighted F1 on that corpus. And a character-level perturbation that drives DeBERTa from 0.9972 to 0.3644 weighted F1 looks like a targeted vulnerability but is not distinguishable from one, because the same model labels uniform random character strings as human in 100% of cases at 0.98 mean confidence. We report the methodological requirement rather than the vulnerability.

Our contributions are four. The first is the decomposition itself, stated formally in Section III-D as a reusable measurement over any human-versus-machine corpus, with the length control that stops its two arms sharing a channel. The second is its application to two widely used benchmarks, with a complete eight-configuration model comparison, cross-corpus transfer over three seeds, and a per-feature SHAP attribution that locates which orthographic property carries the asymmetry. The third is a set of three controls, on tokenisation, pipeline execution and label-free inputs, each of which overturns a conclusion the uncontrolled version of this study had reached. The fourth is a paired statistical treatment, exact-binomial McNemar with a paired bootstrap interval, applied uniformly so that no comparison rests on a difference in the fourth decimal place.

II. Related Work

Detection on these benchmarks. Guo et al. [1] introduce HC3 and report a RoBERTa detector at 99.82 F1, establishing both the corpus and the accuracy regime the field has worked in since. Su et al. [2] show that regime is narrower than it appears, since semantic-invariant tasks drop accuracy to roughly 91.7%. Zhou [3] reaches competitive DAIGT V2 accuracy with classical models over character n-grams and no transformer, a result consistent with ours but reported without the surface-content separation that would explain it and without matching the text budget the transformers were given. Kubrusly et al. [9] likewise report strong bag-of-words results on HC3 without isolating what those features encode, and Ansary [4] and Lamsiyah et al. [10] report near-ceiling DAIGT-family results in-distribution only. Annepaka et al. [11] mix linguistic features with a transformer for 99.45% on HC3, combining the two sources our decomposition separates. Yadagiri et al. [12] pair DeBERTa with auxiliary stylometric features, the closest prior design to ours in ingredients and the furthest in purpose, since it combines the two channels to raise a score where we hold them apart to measure one. Socolof et al. [13] detect with style embeddings alone, a motivation we share, though a learned embedding cannot be shown to exclude content the way a hand-built orthographic vector can. Surveys of the area [14] catalogue detector families without asking what a given corpus is separable by.

Competition solutions on DAIGT V2. The corpus was assembled for a public competition, and the solutions it produced are informative about what the corpus rewards. The winning entry [15] and strong public ones [16], [17] lean on character n-grams and ensembles of classical models rather than on larger language models, and independent replications reach the same place [18]. That pattern is usually read as evidence that classical models suffice for this task. Our decomposition suggests a second reading, compatible with it and easier to act on, namely that a corpus whose surface arm alone reaches 0.9214 rewards models good at surface statistics, so a leaderboard is partly a measurement of the corpus rather than of the method.

Documented artefacts in these corpora. Tian et al. [5] identify the HC3 whitespace convention, observe that a detector could score well by exploiting it, and release a cleaning kit. Their contribution is the artefact and the remedy. They do not quantify how much of the corpus's total separability surface form carries, and they do not test whether a given detector can represent the cue at all, which is the question our tokenisation control answers and which turns out to change the conclusion for BERT. Baidya et al. [6] document a length confound in HC3 and control it by length-matching, again one artefact in isolation, and it is the confound our two arms turn out to share and that Section IV-E closes. Ardeshirifar [7] standardises punctuation before comparing handcrafted features to deep models without measuring what that removes, and Park et al. [8] show HC3-trained detectors rely on prompt-specific collection shortcuts. Where these identify individual cues, we measure the aggregate, which is what a practitioner comparing benchmarks needs.

Shortcut learning and dataset artefacts elsewhere. The pattern is not specific to text detection. Torralba and Efros [19] showed vision datasets are identifiable from their own images, so a classifier can learn the collection rather than the category. In natural language inference, Gururangan et al. [20] and Poliak et al. [21] found a model reading only the hypothesis reaches far above chance, and Niven and Kao [22] traced apparent argument comprehension to lexical cues that vanish under an adversarial split, with Geirhos et al. [23] giving the general account. Our surface-only arm is a hypothesis-only baseline transferred to this task, playing the same role of establishing how far a model gets without the capability the benchmark is taken to measure. What differs is that a hypothesis-only baseline removes a span of text while ours removes a property of it.

Robustness and its controls. Antoun et al. [24] attack HC3-trained detectors with misspellings and homoglyphs and report a fall from 99.88 F1 to 33.57% accuracy, and Huang et al. [25] report comparable degradation under character perturbation. Neither reports a label-free control, and our results suggest such collapses cannot be separated from a generic degenerate response to unreadable input without one, since our own models label random character strings as human with near-maximal confidence. This is a methodological gap rather than a dispute about their measurements. Lai et al. [26] show ensembling preserves in-distribution accuracy while still degrading under shift, Zhang et al. [27] reframe detection as a two-sample test, and Borile and Abrate [28] ablate roughly twenty feed-forward neurons for up to 6.9 points of out-of-distribution gain, evidence that some in-distribution performance rests on localised corpus-specific features.

Fairness of the resulting decisions. Liang et al. [29] show that several deployed detectors misclassify writing by non-native English speakers as machine-generated at high rates, and that trivial rewriting removes the effect. That result bears directly on ours. A detector whose separability comes substantially from orthographic regularity is a detector whose errors will concentrate on writers whose orthographic habits differ from the corpus it was fitted to. We do not measure that effect here, and we note the connection because it is the most consequential reason to care which arm a benchmark's score is coming from.

Cross-corpus evaluation. Alikhanov et al. [30] merge HC3 and DAIGT V2 into one 124,195-sample corpus under a topic-held-out split, reporting 82.87% to 88.86% accuracy against the near-ceiling numbers usual in-distribution. Their design asks whether one model can serve both corpora, ours asks what each corpus is separable by, so the two sit alongside rather than replacing one another. Lu et al. [31] show detectors can be evaded by prompting alone, a threat model orthogonal to the question here.

III. Method

A. Corpora

DAIGT V2 [32] contains 44,868 argumentative student essays, 27,371 human-written and 17,497 machine-generated by a mixture of 2023-era systems including GPT-3.5, Llama-2, Mistral-7B and Falcon-180B, assembled for a public detection competition [33]. The human side derives from the PERSUADE corpus of student argumentative writing [34]. HC3 [1] contains question-answer pairs from five English sources, contrasting human answers with a single generator, GPT-3.5-Turbo-0301, over 85,449 rows [35]. The two make a useful pair because they differ in nearly every respect that matters here. DAIGT V2 has many generators, one genre and long documents. HC3 has one generator, five source domains and short ones. Both were class-balanced by downsampling the majority class, giving 34,994 and 53,806 rows. Balancing fixes the score of a degenerate all-one-class predictor at 0.333 weighted F1 rather than at something respectable, which is what makes the label-free control in Section IV-H readable. Training partitions are balanced to within 0.002 of parity.

Table I gives the duplication audit. DAIGT V2 is essentially free of internal duplication. HC3 carries 6,118 duplicate rows in 5,986 groups, plus 483 rows that are collection artefacts, typically stored API failure messages.

B. Partitioning

That duplication is why the split is group-aware. Documents are grouped by an MD5 hash of their whitespace-normalised, lowercased text and whole groups go to one partition, so no two documents with identical normalised content land on opposite sides of a boundary. The partition is 72/8/20 rather than 80/10/10, because a fifth is taken for test first and a tenth of the remainder for validation. Measured directly, the group-aware rule leaks 0 of 10,732 HC3 test documents while a plain stratified split of the same sample leaks 570, which is 5.30%. DAIGT V2 leaks nothing either way, so no number here is inflated by memorised duplicates. The split is built once and reused unchanged by every model and seed, so a comparison between two models is never also a comparison between two evaluation sets.

TABLE I
CORPUS STATISTICS AFTER BALANCING. DUPLICATION IS MEASURED OVER THE UNBALANCED CORPUS BY EXACT MATCH ON WHITESPACE-NORMALISED, LOWERCASED TEXT. LEAKAGE IS THE SHARE OF TEST DOCUMENTS WHOSE EXACT TEXT ALSO APPEARS IN TRAINING OR VALIDATION, MEASURED ON THE BALANCED SAMPLE UNDER EACH SPLIT RULE. THE GROUP-AWARE RULE IS THE ONE EVERY NUMBER IN THIS PAPER USES.

	DAIGT V2	HC3
Rows, as published	44,868	85,449
Rows, balanced	34,994	53,806
Train / validation / test	25,196 2,800 6,998	38,785 4,289 10,732
Duplicate rows	6 (0.01%)	6,118 (7.16%)
Duplicate groups	6	5,986
Cross-label duplicate texts	0	0
Collection artefact rows	none found	483
Test leakage, group-aware split	0 (0.00%)	0 (0.00%)
Test leakage, naive stratified	0 (0.00%)	570 (5.30%)

C. Models and configurations

Five model families are evaluated. Three are classical, Naive Bayes, logistic regression and a linear support vector machine, each run under bag-of-words and TF-IDF for six classical configurations per dataset. Earlier work at full scale reported only the winning representation per model, so we report the complete grid. Two are transformers, `bert-base-uncased` [36] and `microsoft/deberta-v3-base` [37], fine-tuned with AdamW over a grid of learning rate in $\{2e{-}5, 3e{-}5\}$, batch size in $\{16, 32\}$ and weight decay in $\{0.01, 0.1\}$, which is sixteen configurations per dataset, with the operating point selected on validation weighted F1 and early stopping at patience two.

DeBERTa is the strongest encoder of its size on general language-understanding benchmarks, ahead of both BERT and RoBERTa at comparable parameter counts [37], which is reason enough to include it. Two properties matter more for this study specifically. Its disentangled attention separates content from position, the relevant inductive bias when the question is how much of a corpus is separable by form, and its SentencePiece tokeniser [38] encodes leading whitespace whereas BERT's WordPiece does not, which makes the pair a controlled contrast on exactly the cue Section IV-G examines. Benchmark standing does not by itself predict the outcome here, since the two models are level on DAIGT V2 and separate only on HC3.

Classical models receive lowercased, stopword-filtered, lemmatised text with every non-Latin character removed. Transformers receive raw text, whitespace-normalised only, truncated to 128 tokens. The asymmetry is deliberate, since stripping punctuation and casing removes signal a pretrained transformer can use, but it also confounds any classical-versustransformer comparison, and Section IV-B measures that rather than assuming it away.

D. The surface-content decomposition

The central instrument of this paper separates how much of a benchmark's human-versus-machine separability is available from surface form alone and how much from content alone. Write x for a document and $y \in \{0, 1\}$ for its label, with 1 denoting machine-generated. Let $\mathcal{D}_{\text{tr}}$ and $\mathcal{D}_{\text{te}}$ be the fixed training and test partitions. Two maps are defined on the raw text. The surface map ϕ_{S} sends a document to $\mathbb{R}^{47}$, a vector of orthographic statistics described in Section III-F, and reads no word identity whatsoever. The content map ϕ_{C} first applies the normalisation

$$\nu(x) = \text{collapse}\Big(\text{map}\big(\text{lower}(x),\, \sigma\big)\Big), \tag{1}$$

where lower lowercases, map applies the character rule σ that keeps a character when it is a lowercase Latin letter or whitespace and replaces it with a space otherwise, and collapse reduces runs of whitespace to a single space. The normalised text is then sent to raw bag-of-words counts over the training vocabulary. Punctuation, casing and non-ASCII characters cannot survive ν, so they cannot enter the content arm. Word identity never enters ϕ_{S}.

Each arm fits one logistic regression on the same training partition,

$$h_a = \arg\min_h \sum_{(x,y)\in\mathcal{D}_{\text{tr}}} \ell\big(h(\phi_a(x)), y\big) + \tfrac{1}{2C}\|h\|_2^2, \tag{2}$$

for $a \in \{\text{S}, \text{C}\}$, with ℓ the logistic loss and C fixed at the library default for both arms. Sharing the classifier family and the regularisation strength is what makes the two arms comparable, since any difference between them is then a difference in what the representation carries rather than in how it is fitted.

The quantity we report is the error of each arm on the held-out partition,

$$\varepsilon_a = \frac{1}{|\mathcal{D}_{\text{te}}|} \sum_{(x,y)\in\mathcal{D}_{\text{te}}} \mathbf{1}\big[h_a(\phi_a(x)) \neq y\big], \tag{3}$$

and the separability gap between them,

$$\Delta = \varepsilon_{\text{S}} - \varepsilon_{\text{C}}. \tag{4}$$

A gap near zero says orthography alone reaches the same accuracy as content alone on that corpus. A large positive gap says content carries what orthography does not. The surface arm's own error ε_{S} answers a second and more direct question, namely how much of the benchmark a detector could pass without reading language at all.

Two properties of Δ should be stated plainly. It compares two specific arms rather than partitioning the total separability, since a richer surface featurisation would lower ε_{S} and a stronger content model would lower ε_{C}, so it is a corpus-level comparison under a fixed pair of instruments. It is also not a claim that either corpus is clean, since ε_{S} is far below chance on both.

The *full* arm is the fine-tuned transformer on raw text. It differs from the other two in model family, in parameter count and in the amount of text it reads, so it is reported as a reference point and we draw no conclusion from a transformer-to-classical difference alone without first matching the text budget.

E. The length channel and how it is closed

One channel survives into both arms as defined above. The surface arm reads character, word and sentence counts directly, and raw bag-of-words rows sum to document length, so ϕ_{C} can recover it too. Length is a documented confound on both corpora [6], so we close it rather than assume it away.

On the surface side we drop the five features whose magnitude scales with document size, leaving 42 rates and ratios, and write ϕ_{S}^{-L} for the reduced map. Every dropped feature has a rate twin in the same vector, so the cue itself is not removed along with the length. On the content side we normalise each row to unit total and then rescale by the mean training document length $\bar{n}$,

$$\phi_{\text{C}}^{-L}(x) = \bar{n} \cdot \frac{\phi_{\text{C}}(x)}{\|\phi_{\text{C}}(x)\|_1}. \tag{5}$$

The rescaling is not cosmetic. Rows that sum to one shrink every feature by roughly $\bar{n}$, so at a fixed C the effective penalty becomes far heavier and the arm collapses for reasons of scale rather than of length. We report both variants in Section IV-E and read only the rescaled one, and we treat the unrescaled arm as an instrument artefact rather than a result.

We also fit a *length-only* arm on the three pure document-size features, so the strength of the shared channel is measured directly instead of being inferred from what removing it costs the other two.

F. The surface feature set

TABLE II
THE 47 SURFACE FEATURES, BY FAMILY. NONE READS WORD IDENTITY. THE FIVE FEATURES MARKED AS SIZE-SCALING ARE THE ONES THE LENGTH CONTROL REMOVES, AND EACH HAS A RATE TWIN THAT REMAINS. TWO FEATURES ARE CONSTANT ON THE DAIGT V2 TEST SPLIT AND ONE ON HC3, WHICH IS RECORDED BECAUSE A FEATURE WITH NO VARIANCE CONTRIBUTES NOTHING AND SHOULD NOT BE COUNTED AS EVIDENCE OF COVERAGE.

Family	Count	Notes
Space before punctuation	2	raw count is size-scaling
Document size	3	characters, words, sentences
Word shape	1	mean word length
Casing	3	uppercase, capitalised, all-caps
Non-ASCII and emoji	2	emoji count is size-scaling
Digits	1	rate per 1,000 characters
Whitespace layout	2	double spaces, newlines
Sentence structure	1	mean sentence length
Per-character punctuation rates	32	one per punctuation mark
Total	47	5 size-scaling, 42 rates and ratios

Table II lists the families. Every feature is a count or a rate computable in one pass over the raw string, with no model, no lexicon and no language identification, which is what makes

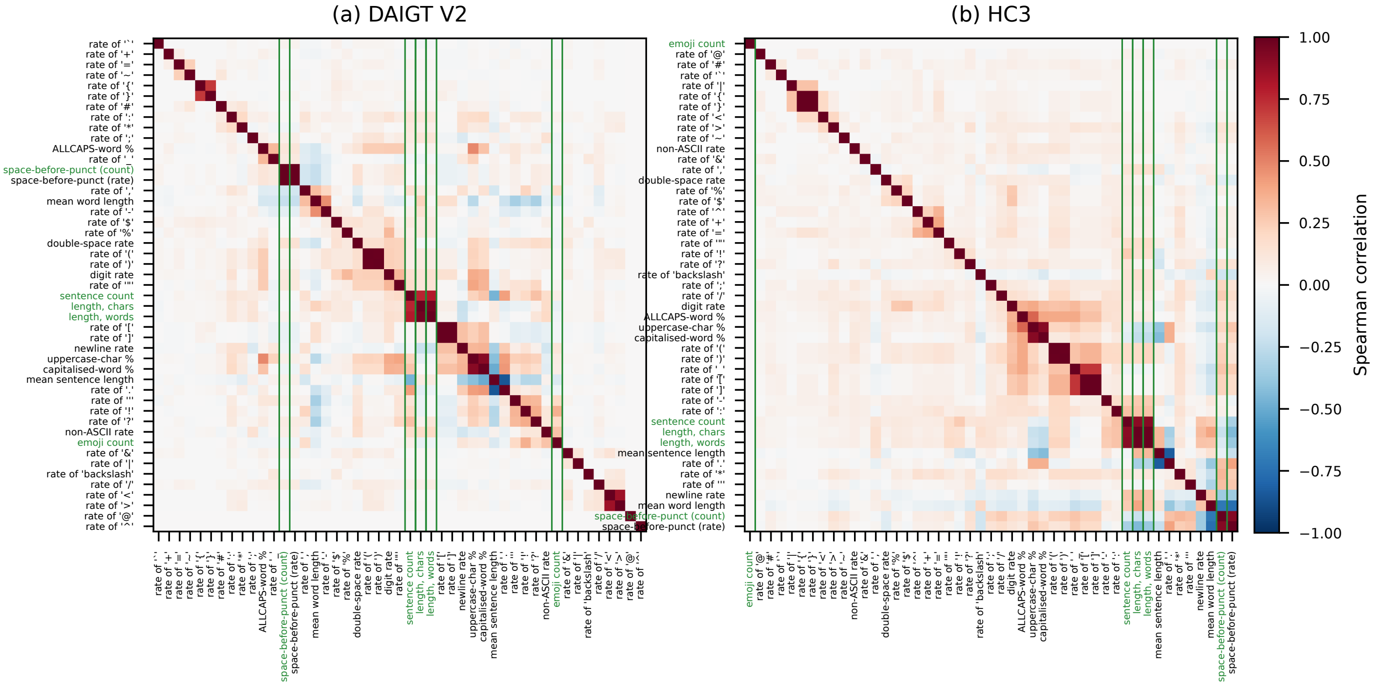

Fig. 1. Spearman correlation among the 47 surface features on each test split, with rows and columns ordered by average-linkage clustering on $1-|\rho|$ so correlated blocks are adjacent. Features the length control removes are outlined and labelled in green. The set is close to unredundant on both corpora, with mean absolute off-diagonal correlation 0.062 on DAIGT V2 and 0.084 on HC3, and only 10 of 1,081 pairs on DAIGT V2 and 14 of 1,081 on HC3 exceeding 0.5 in absolute value. The visible blocks are the ones a reader would predict, namely the bracket and parenthesis rates moving together and the document-size features moving with the sentence count. Features with no variance on a test split have undefined correlation and are drawn as zero.

the measurement cheap enough to run on any new corpus before committing to it.

Fig. 1 shows how much these features overlap. They are close to unredundant, which matters twice over. The surface arm's performance is not one cue counted 47 times, and the length control removes a small, identifiable part of the vector rather than most of it in disguise.

G. Metrics

We report weighted F1, which is what prior work on these corpora reports, alongside error rate, because F1 compresses at this ceiling and a change from 0.9916 to 0.9972 reads as small while the corresponding error rate falls by a factor of three. For class c with precision P_c and recall R_c, support n_c and $N=\sum_c n_c$,

$$\mathrm{F1}_c = \frac{2P_cR_c}{P_c+R_c}, \quad \mathrm{F1}_{\mathrm{w}} = \sum_c \frac{n_c}{N}\mathrm{F1}_c, \quad \mathrm{F1}_{\mathrm{m}} = \frac{1}{|\mathcal{C}|}\sum_c \mathrm{F1}_c. \tag{6}$$

Weighted and macro F1 coincide to four decimal places in every row we report, which is the check that no result here is produced by class skew. We also report ROC area, because a threshold-free ranking measure can disagree with a thresholded one, and on DAIGT V2 it does.

H. Paired statistics

Every transformer baseline is read from the deployed checkpoint's own record rather than from the maximum over its hyperparameter grid, because an earlier version of this analysis compared a grid maximum against a fixed configuration and reported differences that did not exist. Seed spread over three seeds is reported as a range and only for the cell it was measured on, never borrowed across datasets or model families, since fine-tuning outcomes vary materially with initialisation and data order [39], [40]. A range over three runs is not a test statistic, so no comparison here rests on one, and we run no parametric tests at $n \le 30$.

Where two models are compared on the same test set, the comparison is paired, because the two predictions are made on the same documents and treating them as independent samples discards that structure. Let b be the number of documents the first model classifies correctly and the second does not, and c the reverse. Under the null that the two have equal error, the discordant pairs split evenly, so

$$b \mid (b+c) \sim \mathrm{Binomial}\big(b+c, \tfrac{1}{2}\big)\,, \tag{7}$$

and we report the exact two-sided binomial p-value [41] rather than the chi-squared approximation, which is unreliable when $b+c$ is small, as it is for several comparisons here.

An exact test reports whether a difference is distinguishable from zero, not how large it is, so each comparison also carries a paired bootstrap interval on the error difference [42]. Writing $e_i^{(1)}$ and $e_i^{(2)}$ for the two models' per-document error indicators, we resample document indices with replacement $B = 10{,}000$ times and take the empirical 2.5th and 97.5th percentiles of

$$\hat{\Delta}^{(b)} = \frac{1}{n}\sum_{i\in I_b}\Big(e_i^{(1)} - e_i^{(2)}\Big), \tag{8}$$

where I_b is the bth resample. Resampling index sets rather than each model's errors independently is what preserves the pairing.

A bootstrap resamples one test set and therefore says nothing about how the corpus was partitioned in the first place. Because the central claim of this paper is a null on HC3, and a null is exactly the kind of claim a single lucky partition can manufacture, the whole decomposition is repeated over five group-aware partitions of the same balanced sample, varying only the split seed. Section IV-F reports what that shows.

I. Controls

Three controls guard the claims below, and each was added after an uncontrolled version of the corresponding experiment produced a conclusion that did not survive. The *pipeline-fires* control addresses the fact that our whitespace cleaning produces bit-identical BERT predictions on HC3, which taken alone is indistinguishable from a cleaning step that never ran. We apply the same pipeline to DAIGT V2, where it also removes emoji that WordPiece represents, and confirm predictions there do change. The *tokenisation* control compares token identifier sequences for minimally different strings, since whether a detector can exploit a character-level cue is a property of its tokeniser and is directly testable rather than a matter of inference. The *label-free* control scores 400 inputs per condition that carry no human-versus-machine information at all, including uniform random character strings, token-shuffled text, punctuation-only strings, a repeated token and the empty string. Without it, a confident prediction on corrupted input is indistinguishable from a learned response to it, which is the distinction Section IV-H turns on.

IV. Results

A. A classical model matches a transformer on DAIGT V2 but not on HC3

TABLE III

ALL MODEL CONFIGURATIONS, BOTH DATASETS. TRANSFORMER ROWS ARE THE DEPLOYED CHECKPOINTS, NEVER A GRID MAXIMUM, AND CLASSICAL ROWS ARE REFITS ON THE SAME SPLIT. ERROR RATE IS GIVEN BESIDE F1 BECAUSE F1 COMPRESSES AT THIS CEILING. BOLD MARKS THE BEST CLASSICAL AND THE BEST OVERALL ENTRY PER DATASET COLUMN. MACRO F1 EQUALS WEIGHTED F1 TO FOUR DECIMALS IN EVERY ROW, SO NO RESULT IS PRODUCED BY CLASS SKEW.

		DAIGT V2		HC3	
Model	Rep.	F1	err %	F1	err %
Naive Bayes	BoW	0.9591	4.09	0.8713	12.87
Naive Bayes	TF-IDF	0.9577	4.23	0.8670	13.28
Logistic Reg.	BoW	0.9893	1.07	**0.9551**	4.49
Logistic Reg.	TF-IDF	0.9857	1.43	0.9365	6.35
SVM	BoW	0.9870	1.30	0.9475	5.25
SVM	TF-IDF	**0.9910**	0.90	0.9449	5.51
BERT	raw	0.9916	0.84	0.9916	0.84
DeBERTa	raw	**0.9917**	0.83	**0.9972**	0.28

Table III reports every configuration on both datasets. The two benchmarks differ in a way no single headline number reveals. On DAIGT V2 the best classical configuration, an SVM over TF-IDF at 0.9910 weighted F1, sits 0.0007 below the best transformer at 0.9917. On the same test documents the two disagree on 99 cases, 47 to 52, so the exact McNemar test returns $p = 0.69$ and the paired bootstrap puts the error difference at $+0.07$ percentage points with a 95% interval of $[-0.21, +0.34]$ that contains zero. On HC3 the same comparison is 0.9551 against 0.9972, with 484 discordant cases split 16 to 468, $p < 10^{-6}$, and an error difference of $+4.21$ points with interval $[+3.82, +4.61]$. The error-rate ratio there is sixteen to one.

Fig. 2 gives the corresponding ROC curves in the region that matters for a deployed detector, where false positives are expensive because they are accusations.

Ranking by area does not reproduce ranking by F1. On DAIGT V2, BERT attains the higher area, 0.999641 against 0.999144, while DeBERTa attains the higher F1. Neither ordering is real, since the two models disagree on only 75 documents, 37 to 38, at $p = 1.00$. On HC3 the pair separates cleanly at $p < 10^{-6}$. We report both measures for this reason, and we would draw no conclusion from either ordering on DAIGT V2.

Over the full ROC range, not shown, every curve on both corpora sits close to the corner and the two benchmarks look alike. The low-false-positive view is where they separate, which is itself a reason to prefer it.

B. The DAIGT V2 tie is a text budget, not an architecture result

TABLE IV

CLASSICAL MODELS REFITTED ON THE EXACT CHARACTER SPAN EACH TRANSFORMER'S OWN TOKENISER RETAINED, SO BOTH SIDES READ THE SAME TEXT. THE WINDOW IS DEFINED BY OFFSET MAPPING WITH NO DECODE ROUND-TRIP. HELD TO ONE BUDGET THE TRANSFORMER LEADS ON BOTH CORPORA, AND THE DAIGT V2 TIE IN TABLE III DISAPPEARS. THE TWO TOKENISERS DEFINE NEARLY THE SAME WINDOW AND RETURN THE SAME CONCLUSION.

		chars	best classical err %		transf.
Data	Window	kept	full text	128-tok	err %
DAIGT V2	BERT	0.335	0.90	2.10	0.84
DAIGT V2	DeBERTa	0.341	0.90	2.09	0.83
HC3	BERT	0.746	4.49	5.66	0.84
HC3	DeBERTa	0.756	4.49	5.72	0.28

That DAIGT V2 tie is not a matched comparison, and reading it as a statement about architecture would be wrong. The classical models read the whole document while the transformers read 128 tokens, which is 33.5% of a mean DAIGT V2 essay against 74.6% of a mean HC3 answer. Table IV refits the classical grid on the exact span each checkpoint's own tokeniser kept. The best DAIGT V2 configuration then rises from 0.90% to 2.10% error against BERT's 0.84%, a difference of 1.257 points with interval $[+0.900, +1.629]$ at $p < 10^{-6}$, and the DeBERTa window returns the same 1.257. On HC3 the gap only widens, from 4.49% to 5.66%. Held to one text budget the transformers lead on both corpora.

What the unmatched comparison shows is narrower and still worth stating, namely that a bag-of-words model reading a whole essay matches a transformer reading its opening. That is a claim about information access, not architecture.

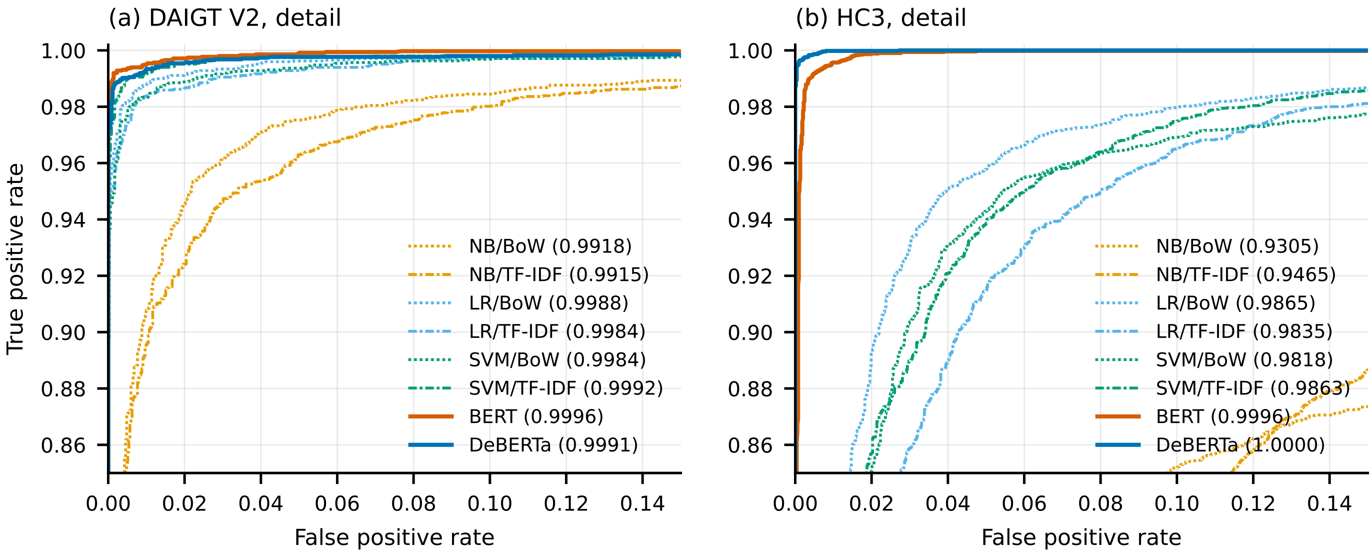

Fig. 2. ROC curves for all eight model configurations, restricted to the low-false-positive region where the models actually differ. Areas under the full curve are given in the legend. On DAIGT V2 in panel (a) every configuration converges toward the corner and the classical models are close behind the transformers. On HC3 in panel (b) the two transformers separate sharply from all six classical configurations, which is the same split Table III shows and the reason a single averaged benchmark score over both corpora would hide it. Colours are colourblind-safe and each series carries a distinct dash pattern, so the ordering survives greyscale reproduction.

C. On HC3, orthography alone matches content alone

TABLE V
SURFACE-CONTENT DECOMPOSITION. SURFACE-ONLY USES 47 ORTHOGRAPHIC FEATURES AND NEVER READS A WORD. CONTENT-ONLY IS UNFILTERED BAG-OF-WORDS AFTER PUNCTUATION, CASING AND NON-ASCII CHARACTERS ARE STRIPPED. THE LOWER BLOCK CLOSES THE DOCUMENT-LENGTH CHANNEL THAT BOTH PRIMARY ARMS CAN OTHERWISE READ. BOLD MARKS THE BETTER ARM IN EACH BLOCK AND DATASET COLUMN. THE TRANSFORMER IS A REFERENCE POINT, NOT A MATCHED ARM.

	DAIGT V2		HC3	
Arm	F1	err %	F1	err %
surface-only	0.9214	7.86	**0.9680**	**3.20**
content-only	**0.9901**	**0.99**	0.9674	3.26
full transformer	0.9917	0.83	0.9972	0.28
length channel closed				
length-only (3 feats)	0.7540	24.59	0.8475	15.23
surface-only, no length	0.9007	9.93	**0.9662**	**3.37**
content-only, length-norm.	**0.9906**	**0.94**	0.9371	6.28

Table V gives the decomposition. On HC3 the two arms are statistically indistinguishable, 3.20% error from orthography against 3.26% from content. They disagree on 625 documents, split 316 to 309, so the exact McNemar test returns $p = 0.81$ and the paired bootstrap puts the difference at -0.07 percentage points with a 95% interval of $[-0.52, +0.40]$. Forty-seven features that never read a word do as well as a bag-of-words model over the whole vocabulary.

On DAIGT V2 the same two arms separate by a factor of 7.9 in error, 7.86% against 0.99%, and there the separation is unambiguous, with 569 discordant documents split 44 to 525, $p < 10^{-6}$, and a difference of $+6.87$ points with interval $[+6.23, +7.52]$.

Surface form is informative on both benchmarks, since DAIGT V2's surface-only arm reaches 0.9214 weighted F1, far above the 0.50 a coin would give and far above the 0.333 a degenerate single-class predictor would give. The finding is not that one corpus is clean and the other is not. It is that only on HC3 does surface rise to parity with content.

The content-only arm reaches 0.9901 on DAIGT V2 against the transformer's 0.9917, within 0.16 points of error, under the same unmatched text budget as Section IV-B. On HC3 the transformer improves on either arm by roughly 0.03 weighted F1, which is consistent with it combining information neither arm holds alone.

D. One orthographic feature carries the HC3 result

The decomposition in Section IV-C says how much orthography carries. It does not say which orthographic property carries it, and the two corpora turn out to differ there as sharply as they differ in the aggregate. Fig. 3 gives per-feature Shapley attributions [43] over the surface arm. Because the arm is a linear model, the attributions are exact rather than sampled, so the ranking carries no estimation error of its own.

On HC3 the single largest contributor is the rate of spaces preceding punctuation, at mean absolute SHAP 7.81 against 3.35 for the next feature, and its fitted coefficient is negative, so its presence drives the human label. That is the collection convention Tian et al. [5] identified, recovered here without being looked for and quantified against the rest of the feature set. Section IV-G then shows why its presence does not license the conclusion a reader would reach next.

On DAIGT V2 the leading contributors are document length at 2.77, mean word length at 1.88 and casing statistics, none of which exceeds 2.8 and none of which is a collection convention in the same sense. The surface arm on that corpus is reading a diffuse stylistic signal rather than a single artefact, which is consistent with its much weaker showing against the content arm.

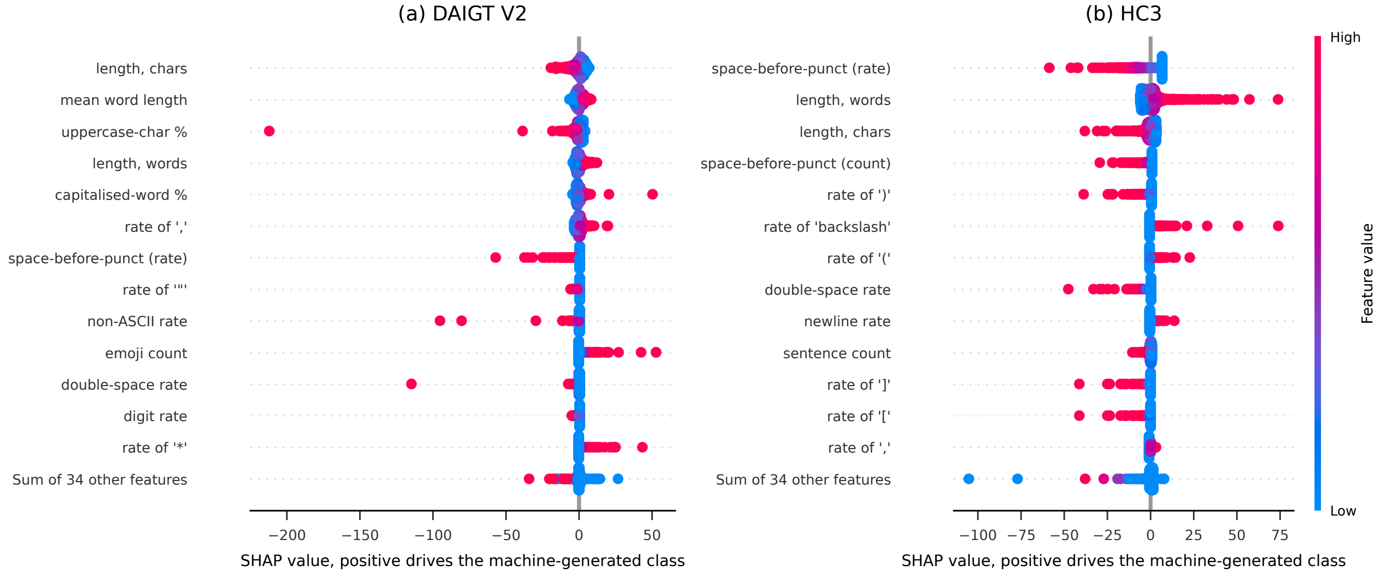

Fig. 3. SHAP attributions over the surface-only arm, computed exactly with a linear explainer on 2,000 held-out documents per corpus. Each point is one document, positioned by its contribution to the machine-generated class and coloured by the feature's value. The two corpora look nothing alike. On HC3 in panel (b) the rate of spaces preceding punctuation dominates every other feature and pushes hard toward the human class. On DAIGT V2 in panel (a) no orthographic feature dominates, and the leading contributors are document length and word shape rather than a single collection convention.

E. Neither result is an artefact of document length

Both arms can read document length, so the comparison above could be a length comparison in disguise. Length alone reaches 24.59% error on DAIGT V2 and 15.23% on HC3, well short of either arm but far from nothing, which is why the channel is closed rather than argued away.

Closing it on both sides sharpens rather than weakens the two conclusions. On DAIGT V2 the length-free content arm loses nothing, 0.94% against 0.99% at $p = 0.71$, while the length-free surface arm rises to 9.93%, widening content's advantage from 7.9 to 10.6 times. On HC3 the surface arm barely moves, from 3.20% to 3.37%, while the content arm rises to 6.28%, so orthography no longer merely matches content but beats it by 2.91 points, with interval $[+2.37, +3.44]$ at $p < 10^{-6}$. Removing length costs the content arm eighteen times what it costs the surface arm, which is the opposite of what a length-driven surface result would produce.

One caution on the instrument belongs here, because it nearly produced a third false conclusion. Normalising bag-of-words rows to unit total without restoring their scale shrinks every feature by roughly the mean document length, so at a fixed regularisation strength the penalty becomes far heavier and the arm collapses to 9.69% on DAIGT V2 and 14.25% on HC3. Read uncritically that looks like a dramatic length effect. It is an artefact of scale, visible in the optimiser trace, since the unrescaled arm converges in 16 to 24 iterations against 121 to 192 for the raw-count arm. Every figure quoted above uses rows rescaled to the raw-count arm's magnitude, and the unrescaled arm is reported only as a negative control.

F. The HC3 null survives repartitioning

A null on one partition is the kind of claim a lucky split can manufacture, so the decomposition is repeated over five group-aware partitions of the same balanced sample. Table VI reports the result, and seed 42 reproduces the single-split numbers exactly, which is the harness self-check that makes the other four readable. Content leads on DAIGT V2 on all five partitions, by 6.00 to 6.87 points, at $p < 10^{-6}$ throughout. On HC3 the surface-content difference is significant on none of the five, with p equal to 0.81, 0.27, 0.34, 0.23 and 0.78, and the difference changes sign between partitions, ranging from -0.30 to $+0.27$ points. That sign instability is the strong form of the null, because an underpowered real difference keeps its sign while a genuine null does not. The length-free comparison, by contrast, is significant on all five and keeps its sign, from -2.67 to -2.91 points, so the advantage orthography gains once length is removed is not a partition effect either.

TABLE VI

The decomposition repeated over five group-aware partitions of the same balanced sample, varying only the split seed. Split variance is between 0.17 and 1.29 percentage points, so the single-split figures in Table V were not misleading, but the parity claim rests on this table rather than on one partition.

Arm	DAIGT V2 mean err % (range)	HC3 mean err % (range)
surface-only	7.15 (6.74–7.86)	3.16 (3.07–3.33)
content-only	0.82 (0.66–0.99)	3.24 (3.06–3.41)
surface-only, no length	9.76 (9.47–9.93)	3.26 (3.14–3.37)
content-only, length-norm.	0.84 (0.77–0.94)	6.02 (5.82–6.28)
length-only	24.31 (23.56–24.85)	15.32 (15.23–15.49)

G. A model that cannot represent the cue reaches 0.9916 anyway

The most-discussed HC3 artefact is a space preceding punctuation in human answers [5], and Section IV-D confirms it is the single strongest orthographic feature on that corpus. In our test split it appears 10.745 times per human document against 0.013 per machine one, and in 88.7% of human documents against 0.3% of machine ones. DAIGT V2 carries the same cue far more weakly, 0.548 against 0.004 per document, in 17.1% against 0.2% of documents.

The natural next inference is that HC3-trained detectors exploit it. Tokenisation makes that testable rather than rhetorical, and the test does not support it. BERT's WordPiece tokeniser splits on punctuation irrespective of adjacent whitespace. For the strings `"the answer is simple ."` and `"the answer is simple."` it emits identical identifier sequences, in three of three pairs tested, while DeBERTa's SentencePiece encodes the leading whitespace and distinguishes all three. BERT cannot represent the cue at all and still reaches 0.9916 weighted F1 on HC3. The cue is therefore sufficient in isolation, since the surface arm that reads it reaches 0.9680, and unnecessary in practice, since a model blind to it reaches 0.9916.

This is also why removing the cue changes nothing for BERT. Applying whitespace cleaning to HC3 yields predictions bit-identical to the uncleaned run, verified by SHA-256 over the saved probability arrays. Taken alone that would be indistinguishable from a cleaning step that never executed. The same pipeline applied to DAIGT V2, where it also removes emoji that WordPiece does represent, does change BERT's predictions. The pipeline fires, and the null is specific to whitespace.

We do not attribute the BERT-to-DeBERTa difference on HC3 to this cue, since the two models differ in tokeniser, architecture, pretraining corpus and parameter count and this experiment separates none of those. What it establishes is narrower and, for anyone planning a cleaning ablation, more useful, namely that a null from removing a cue is uninterpretable without first checking the model could read the cue.

H. The adversarial collapse is not distinguishable from a degenerate response

TABLE VII

LABEL-FREE CONTROL. PERCENTAGE OF INPUTS ASSIGNED THE *human* LABEL, WITH MEAN MAXIMUM SOFTMAX PROBABILITY IN PARENTHESES, ON 400 INPUTS PER CONDITION THAT CARRY NO HUMAN-VERSUS-MACHINE INFORMATION AT ALL. A DETECTOR WITH A CALIBRATED RESPONSE TO UNREADABLE INPUT WOULD NOT ANSWER THESE CONFIDENTLY IN EITHER DIRECTION. ALL FOUR CHECKPOINTS DO.

Checkpoint	rand. chars	shuffled	punct.
DAIGT V2 BERT	0% (0.91)	94.8% (0.99)	97.8% (0.71)
DAIGT V2 DeBERTa	100% (0.99)	100% (1.00)	100% (1.00)
HC3 BERT	51.0% (0.71)	98.5% (1.00)	100% (1.00)
HC3 DeBERTa	100% (0.98)	100% (1.00)	100% (0.99)

Perturbing HC3 test text with character-level typos at 10% of alphabetic characters drives DeBERTa from 0.9972 to 0.3644 weighted F1. The confusion matrix is one-directional, $[[5376, 1], [5207, 148]]$, with human documents still classified almost perfectly and machine documents reassigned to human. On unperturbed data the same checkpoints split their errors roughly evenly between the two directions, so against that baseline this looks like a targeted vulnerability. It is not what the control shows.

Table VII reports the label-free control. On inputs carrying no human-versus-machine information, the same model assigns the human label to uniform random character strings in 100% of cases at mean confidence 0.98, and to token-shuffled and punctuation-only text likewise. All four checkpoints label the latter two conditions human between 94.8% and 100% of the time. The behaviour is specific to neither HC3 nor the attack.

Three candidate mechanisms were tested and none survives. Majority-prior collapse is excluded by the training balance, which is 0.4981 against 0.5019 on DAIGT V2 and 0.4999 against 0.5001 on HC3. Cue injection is excluded because the perturbation does not create the whitespace artefact, moving the machine-side count only from 0.000 to 0.005. A learned association from disfluency to human authorship is unsupported, since subword fertility runs the wrong way, at 1.1192 for HC3 human text against 1.1980 for machine.

We therefore decline to read our own perturbation numbers as robustness measurements. The experiment establishes a requirement on future work rather than a property of these benchmarks.

One contrast within the perturbation results is worth stating even so. Back-translation rewrites the text while leaving it readable and costs all four models between 0.004 and 0.020 weighted F1. Character-level corruption, which makes the text unreadable, is what produces the collapses, with HC3 BERT falling to 0.3746 and HC3 DeBERTa to 0.3644 under 10% typos. That pattern is consistent with the label-free reading and not with a targeted cue-removal reading.

I. Transfer and ensembling

TABLE VIII

CROSS-CORPUS TRANSFER, THREE SEEDS PER CELL. EACH MODEL IS TRAINED ON ONE CORPUS AND EVALUATED ON THE OTHER WITH NO ADAPTATION. RANGES ARE THE OBSERVED MINIMUM AND MAXIMUM OVER THE THREE SEEDS AND ARE NOT CONFIDENCE INTERVALS.

Trained	Model	in-dom.	cross-dom. (range)	gap
DAIGT	BERT	0.9921	0.7902 (0.770–0.810)	0.2019
DAIGT	DeBERTa	0.9930	0.9096 (0.891–0.926)	0.0833
HC3	BERT	0.9930	0.8311 (0.813–0.852)	0.1619
HC3	DeBERTa	0.9970	0.8512 (0.830–0.887)	0.1458

Table VIII reports transfer in both directions over three seeds, with each model trained on one corpus and evaluated on the other without adaptation. Every cell loses between 0.0833 and 0.2019 mean weighted F1, which is what should happen if a substantial part of each corpus's separability is corpus-specific. DeBERTa's gap is roughly half BERT's in both directions at the mean level. We report that as a mean-level pattern that survives three-seed averaging rather than as a confirmed effect, because the observed ranges for BERT's two directions nearly touch and this project runs no parametric tests at three seeds.

A soft-vote ensemble over the two deployed checkpoints, with the mixing weight selected on validation and applied once to test, does not reliably improve on its stronger member. On DAIGT V2 an equal mix reaches 0.9936 against DeBERTa's

0.9917, but the exact McNemar test returns $p = 0.085$ and the bootstrap interval on the error difference is $[-0.39, +0.01]$ percentage points, which contains zero. On HC3 the validation sweep places zero weight on BERT, so the ensemble is DeBERTa and reports DeBERTa's score. We record the degenerate weight rather than reporting that as an ensemble result, because a soft-vote ensemble whose selected weight has discarded one member is not an ensemble. Reporting either row as an improvement would be reading a difference in the fourth decimal place as a finding, and we treat the ensemble as a discussion point, consistent with Lai et al. [26], who find ensembling preserves in-distribution accuracy without addressing the shift that actually degrades these models.

J. A consolidated view

Fig. 4 collects the results above. Panel (c) is where the paper's central claim is either supported or refuted. The DAIGT V2 series sits six to seven points above zero on every partition, and the HC3 series straddles zero and changes sign. Panel (f) shows the same distinction in interval form, where grey marks the intervals containing zero, and those are exactly the comparisons this paper declines to call differences.

V. Discussion

The decomposition separates two benchmarks that a headline accuracy figure does not. On HC3 orthography alone matches content alone and a paired test cannot distinguish them on any of five partitions. On DAIGT V2 content is stronger by a factor of 7.9 in error. A practitioner choosing an evaluation corpus, or a reviewer reading a reported score, cannot recover that distinction from the score itself.

The attribution in Section IV-D says where the difference comes from. HC3's surface arm is close to a single-feature model, while DAIGT V2's reads a diffuse mixture of length, word shape and casing with no comparable leader. Those are different kinds of corpus even though both are separable by form, and the practical consequence differs. A single collection convention can be cleaned. A diffuse stylistic signal cannot, and it may not even be an artefact.

What the measurement does not license is a claim that either corpus is clean. DAIGT V2's surface-only arm reaches 0.9214 weighted F1, so surface form is substantially informative on both. The finding is comparative, that surface reaches parity with content on one corpus and not the other, and it is bounded by the arms we built.

Two results here began as the opposite conclusion, and both say more about method than about these corpora. A cleaning experiment reporting no change is uninterpretable without evidence the pipeline ran, and a tokeniser that cannot represent the cue being cleaned makes the null a fact about the model rather than the corpus. A collapse under perturbation is uninterpretable without a label-free control, since a model that calls random strings human at 0.98 confidence produces the same curve with no vulnerability behind it. We suggest comparable published collapses be read with that control in mind.

Three reporting practices here were adopted after an error reached a draft, and each is cheap. Transformer baselines are read from the deployed checkpoint's own record rather than a grid maximum, after an ablation comparing the two reported differences that did not exist. Seed spread is quoted only for the cell it was measured on, after one was borrowed across corpora as a noise floor. And a leakage rate quoted from an earlier draft had no source file, so it was removed and then measured.

There is a connection to fairness we did not measure and that is the most consequential reason to care which arm a score comes from. Liang et al. [29] show deployed detectors misclassify non-native English writing as machine-generated at high rates. A detector whose separability rests on orthographic regularity should be expected to concentrate its false positives on writers whose orthographic habits differ from the corpus it was fitted to. Our measurement predicts which corpora carry that risk most sharply. Establishing the effect would need author metadata neither corpus provides.

VI. Limitations

Several limits bound every number above. The transformers see at most 128 tokens, so the DAIGT V2 transformer results describe classification from an essay's opening rather than the essay, and the cross-corpus comparison in Table VIII remains not like-for-like in the text each model receives. Seed spread is estimated from three seeds and one such estimate moved by 31% on a rerun, because GPU training here is not bitwise deterministic, so no claim rests on a seed range and every comparison uses a paired test on saved predictions instead. The content arm is a bag-of-words model, so word order and syntax lie outside it and a syntactic content arm could narrow the DAIGT V2 gap. The surface set is 47 hand-built features, two with no variance on the DAIGT V2 test split and one on HC3, so it is not exhaustive and ε_S is an upper bound on surface-arm error. HC3 has one generator collected at one time, both corpora are English, so the orthographic conventions the surface arm reads are not language-independent. We evaluate two corpora and five model families and claim nothing beyond them, in particular not that either corpus represents its genre, only that these two widely used benchmarks differ on the axis we measure.

VII. Conclusion

We proposed a decomposition that measures how much of a machine-generated-text benchmark's separability is available from surface form alone rather than from content, and applied it to DAIGT V2 and HC3 across five model families and eight configurations per corpus. On HC3 the two arms are indistinguishable under paired testing on five partitions, and a per-feature attribution shows a single collection convention accounts for most of the surface arm. On DAIGT V2 content leads by a factor of 7.9 in error and the surface signal is diffuse. Closing the length channel both arms share strengthens both conclusions. Three controls, on tokenisation, pipeline

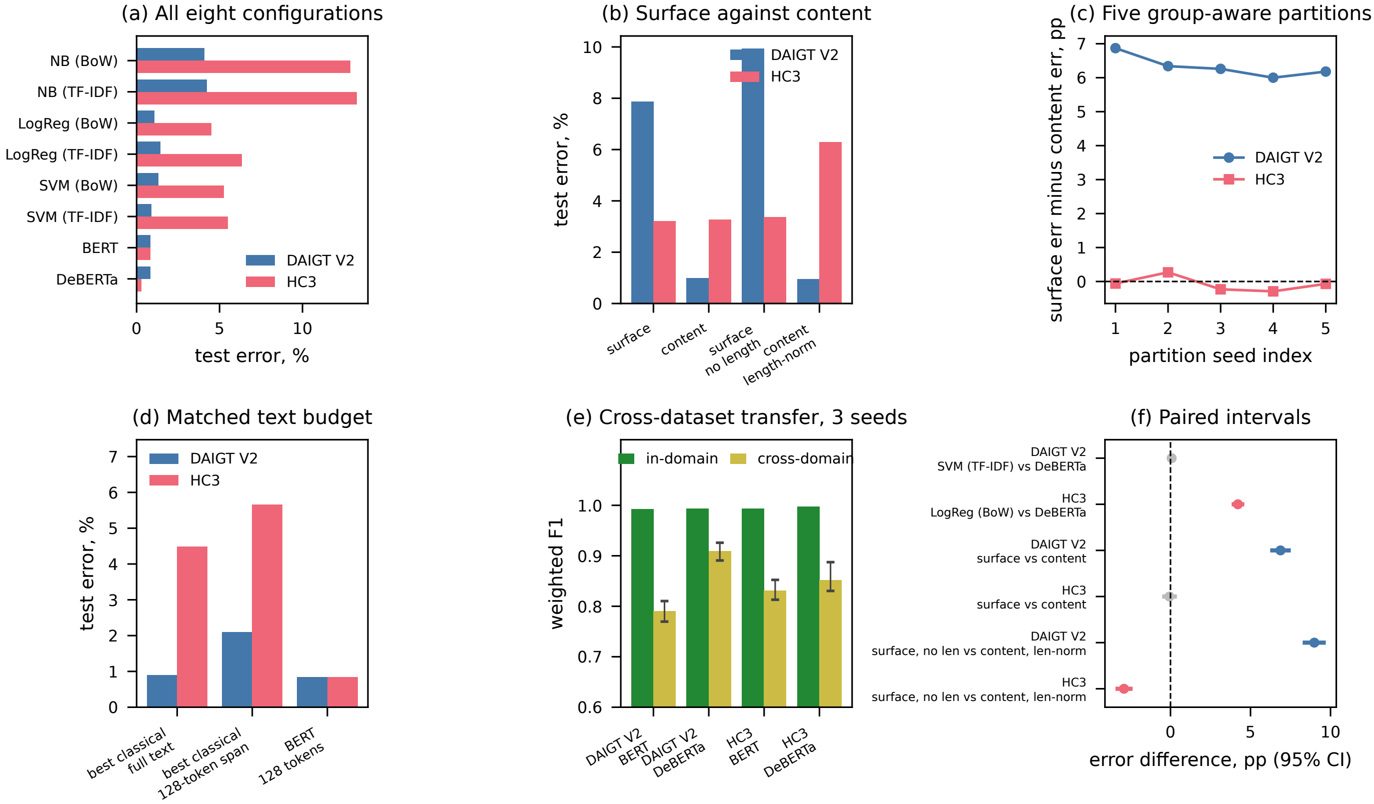

Fig. 4. Every headline result in one view. Panel (a) ranks all eight configurations by test error on both corpora. Panel (b) gives the decomposition with and without the length channel. Panel (c) plots the surface-minus-content error difference across five group-aware partitions, where the DAIGT V2 series sits far above zero on all five and the HC3 series crosses it. Panel (d) shows the matched text budget from Table IV. Panel (e) shows cross-corpus transfer with the observed three-seed range as error bars. Panel (f) shows the paired bootstrap intervals behind the significance claims, with intervals containing zero drawn in grey. Every value is read from the JSON the audit scripts wrote and none is entered by hand.

execution and label-free inputs, each overturned a conclusion this study had reached without them.

The measurement itself is the portable part. It needs no new annotation, it fits two logistic regressions, and it runs in minutes on a laptop. It yields one number per corpus that a headline accuracy cannot express, namely how much of that corpus a model could pass without reading the language. We would encourage reporting it alongside any new detection benchmark, in the same way a hypothesis-only baseline is now reported alongside a natural language inference dataset [21].

Data and Code Availability

Both corpora are public. Every number in this paper is produced by a committed script that writes a JSON file, and no number was transcribed from a session log. The partition is built once by a script that asserts no duplicate-content group crosses a boundary, and is reused unchanged by every model and every training seed. A single notebook reproduces the preprocessing, both transformers at their deployed configurations and the ensemble, reading each configuration from the checkpoint record at runtime rather than from a value typed into the notebook. Code is released for reproduction. The link is withheld for blind review.

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